QP CODE: 25900972

Scheme of Valuation

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER2025

(2024 Admission onwards)

Discipline Specific Core Course (DSC)

MG1DSCSTA100 - FUNDAMENTALS OF STATISTICS AND DATA VISUALISATION

Part A [Answer **any 7** Questions. Each Question carries **2 marks]**

1. Correct definition of scale of measurement or definitions of any two scales of measurements - 2 marks. Or
   Listing down all the 4 scales of measurement - 0.5 each.
2. Any 2 merits of primary data - 1 mark each.
   Definition of primary data only - 1.5 marks.
3. Arrange the data in ascending or descending order - 1 mark,
   Correct answer (= 7) - 1 mark.
   Correct definition of median only - 1 mark.
4. Mentioning the important steps in the use of random number tables for generating random samples - 2 marks.
   Definition of simple random sampling or random number table only - 1.5 marks.
5. Definition or formula of QD - 2 marks.
   Definition of measures of dispersion or quartiles or the statement *'QD is a measure of dispersion'* only - 1 mark.
6. Definition of boxplot or outliers or any valid representation of boxplot with outliers - 2 marks.
7. Definition of measures of central tendency or correct definition of at least 1 measure - 2 marks. Or
   Listing down any 4 measures of central tendency only - 0.5 mark each.
8. Correct reason or definition of stratified sampling - 2 marks.
9. Correlation: strong and positive - 1 mark each.
   Definition or formula of a correlation coefficient (either Pearson or rank) or positive correlation only - 1 mark.
10. Definition or example of relative frequency (fi/N) - 2 marks.
    Definition of frequency table or frequency only - 1.5 marks.

[7x2 = 14 marks]

Part B [Answer **any 4** Questions. Each Question carries **6 marks]**

11. Correct definition or example of less and greater than ogives - 3 marks each.
12. Formula for 3rd and 4th central moments (using any raw moments) - 2 marks each,
    Correct substitution for the results - 1 mark each.
    Correct definition of moments only - 1 mark each up to 4 marks.
13. Estimation of median (Average of 19 and 49 = 34) - 4 marks,
    Mean deviation from median (= 27.2) - 2 marks.
    Formula of median only - 1 mark,

Formula of mean deviation about median only - 2 marks.

14. Sum of squared deviations of rank (= 26) - 3 marks,
    Estimation of rank correlation (= 0.6905) - 3 marks.
    Formula of rank correlation only - 3 marks.
15. (i) Sample space - 2 marks,
    Event definitions - 1 mark each.
    Definition of partition of sample space only - 2 marks
    (ii) {1, 3, 5} - 2 marks.
16. Correct statement of the problem and expansion of the statement using the definition of conditional probability or Bayes theorem - 3 marks,
    - Let R denotes selection of a red pair, H denotes selection of a pair with holes and N denotes selection of a pair without holes then we required P(H|R).
    - Using the definition of conditional probability, $P(H|R) = \frac{P(R \cap H)}{P(R)}$.
    - Using the Bayes theorem $P(H|R) = \frac{P(H)\,P(R|H)}{P(H)\,P(R|H) + P(N)\,P(R|N)}$.

    Estimation of elementary probabilities, correct substitution and answer (= 2/5) - 3 marks,
    - Using conditional probability $\frac{2/15}{5/15}$
    - Using Bayes theorem $\frac{3/15 \times 2/3}{3/15 \times 2/3 + 12/15 \times 3/12}$

    Definition of conditional probability or statement of Bayes theorem only - 4 marks.

[4x6 = 24 marks]

Part C [Answer **any 1** Question. Each Question carries **12 marks]**

17. Definition of skewness and kurtosis - 4 marks each,
    Any two measures of skewness and kurtosis - 1 mark for each measure.
    Diagrammatic representation of skewness and kurtosis only - 2.5 marks each.
18. Solution of the equations and Identification of means [$\bar{X} = 5$, $\bar{Y} = 10$] - 6 marks,
    Identification of the regression equations: regression equation of y on x (8x-10y+60 = 0) and regression equation of x on y (40x-18y-20 = 0) - 2 marks,
    Regression coefficients [byx = 8/10, bxy = 18/40]- 2 marks,
    Estimation of $r^2 = 144/400$ and hence r = 12/20 - 2 marks.
    Theory behind the steps only - 2 marks each up to 6 marks.

[1x12 = 12 marks]